National University of Computer and Emerging Sciences

Lab Manual 02

Programming Fundamentals

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Semester	Fall 2025

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Activities:

Activity 1:

Installation of Dev-C++ and Microsoft Visual Studio 2013 is demonstrated by instructor.

Activity 2:

Use of Dev-C++ and Microsoft Visual Studio 2013 is demonstrated by instructor.

Activity 3:

Following basic programs are demonstrated by instructor.

- Write a program that displays ***Hello World!*** on console.

```
#include <iostream>
using namespace std;

int main()
{
	cout << "Hello World!";

	return 0;
}
```

- Write a program that will demonstrate the use of escape sequences as well as endl.

```
#include <iostream>
using namespace std;

int main()
{
	cout << "Hello World!";
	cout << "Hello World!";
	cout << endl;

	cout << "Hello World!\n";
	cout << "Hello World!\n\n";
	cout << "Hello World!\n";
	cout << endl;

	cout << "Hello\bWorld!\n";
	cout << "Hello\b\bWorld!\n";
	cout << "Hello\b\bW\n";

	cout << "Hello\tWorld!\n";
	cout << "Hello\t\nWorld!" << endl;
	cout << "Hello\n\tWorld\n!" << endl;


	cout << "\"Hello World!\".\n";
	cout << "\'Hello World!\'" << endl;
	cout << "Hello\rWorld!" << endl;
	cout << "\"D:\\PF\\Lab 1\\Activity 4.cpp\"\n";

	return 0;
}
```

- Write a program that demonstrate the type casting.

```
#include <iostream>
using namespace std;

int main()
{
```

```
    cout << "7 / 2 = " << 7 / 2 << endl;
    cout << "7.0 / 2 = " << 7.0 / 2 << endl;
    cout << "7 / 2.0 = " << 7 / 2.0 << endl;
    cout << "7.0 / 2.0 = " << 7.0 / 2.0 << endl << endl;

    cout << "(float)7 / 2 = " << (float)7 / 2 << endl;
    cout << "7 / (float)2 = " << 7 / (float)2 << endl;
    cout << "(float)7 / (float)2 = " << (float)7 / (float)2 << endl;
    cout << "(float)(7 / 2) = " << (float)(7 / 2) << endl;
    cout << "(int)7.0 % (int)2.0 = " << (int)7.0 % (int)2.0 << endl << endl;

    cout << "4.0 + 3 * 7 / 2 = " << 4.0 + 3 * 7 / 2 << endl;
    cout << "4 + 3 * 7.0 / 2 = " << 4 + 3 * 7.0 / 2 << endl;

    return 0;
}
```

Activity 4:

Write the program for the followings:

1. Given the radius in inches, and price of a pizza as an input, write a program to find the price of the pizza per square inch.
2. Write a program that calculates the amount of paint required to paint a wall. The program should take the height and width of the wall (in feet) and the coverage of the paint (in square feet per gallon) as inputs and output the number of gallons needed.
3. To make a profit, the prices of the items sold in a furniture store are marked up by 80%. After marking up the prices each item is put on sale at a discount of 10%. Write a program to find the selling price of an item sold at the furniture store. What information do you need to find the selling price?
4. Write a program that helps the user manage their personal budget. The program should prompt the user to input their monthly income, and then ask for the total amount they spend on rent, groceries, utilities, transportation, entertainment, and other expenses. Calculate and display the total expenses, the remaining balance, and the percentage of income spent in each category.
5. Write a program that input two variables, swap the values of those two variables using third variable. Also, output the final values of those variables.
6. Write a program that input two variables, swap the values of those two variables without using third variable. Also, output the final values of those variables.